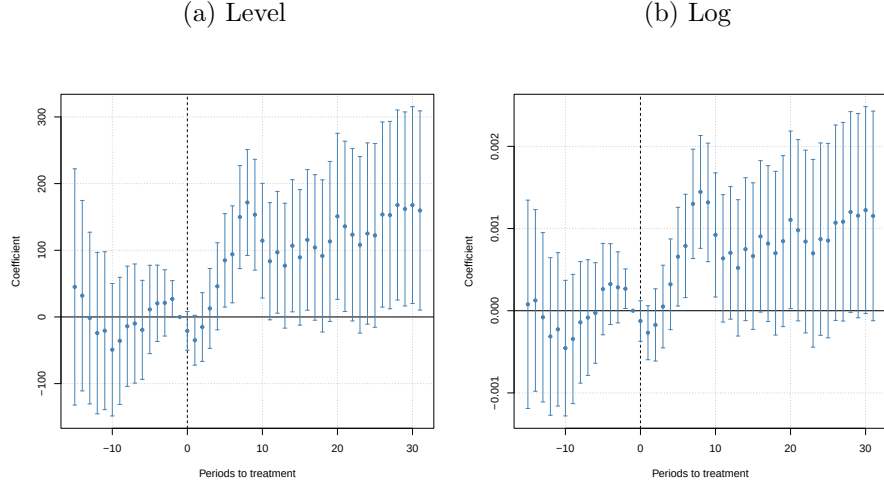


Figure 3: Effect of LLM exposure on firm-level compensation



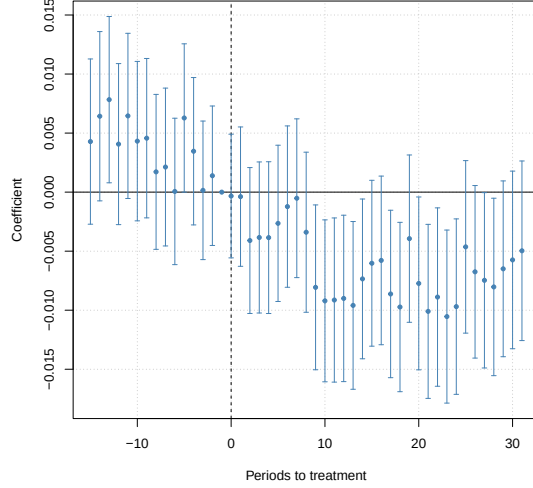
Notes: This figure plots event study estimates of the effect of LLM exposure on average compensation within UK firms. Compensation is defined as annual base salary plus bonuses. Panel A shows compensation in levels, Panel B in logs. Coefficients are estimated from Equation (3) and are shown relative to October 2022, the final pre-treatment month. All regressions include firm fixed effects and industry-by-period fixed effects. Standard errors are clustered at the firm level. The vertical line marks November 2022, when ChatGPT was publicly released. Shaded areas denote 95% confidence intervals.

Figure 4 presents event study estimates for the probability that exposed vs. less exposed firms post any job vacancy in a given month. The results show a change in hiring patterns following November 2022. Prior to this date, firms with different exposure levels did not have statistically significantly different trends in job postings.⁸ Following the release of ChatGPT, however, we observe that a one standard deviation increase in LLM exposure reduces firms' likelihood of posting new jobs by 1.0 percentage point ($p < 0.001$, see Table A1). This effect emerges gradually and becomes statistically significant after approximately 10 months, mirroring the timeline of employment stock adjustments.

Next, we decompose the aggregate hiring reduction by estimating occupation-specific effects across job categories. We estimate Equation (4) separately for 50 distinct occupational categories, with the outcome variable indicating whether a

⁸Although all pre-trend estimates are statistically insignificant, the point estimates exhibit a slight downward drift. If this reflects a differential pre-trend rather than sampling variation, the post-treatment estimates may overstate the true effect.

Figure 4: Effect of LLM exposure on firm-level job listings

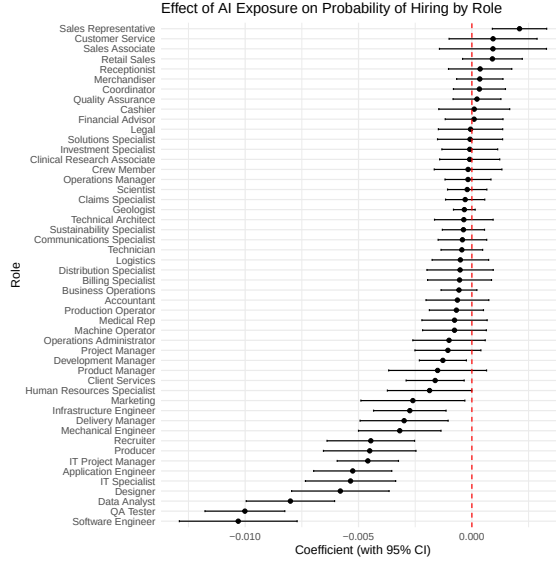


Notes: This figure plots event study estimates of the effect of LLM exposure on the probability that a UK firm posts any job vacancy in a given month. Coefficients are estimated from Equation (3) with firm fixed effects and industry-by-period fixed effects. Standard errors are clustered at the firm level. The dependent variable is an indicator equal to one if the firm posted at least one vacancy in month t . The vertical line marks November 2022, when ChatGPT was publicly released. Shaded areas denote 95% confidence intervals.

firm posts any vacancy for that specific role in a given month. We continue to compare exposed vs. less exposed firms in the same time period.

Figure 5 presents these occupation-specific treatment effects, ordered by magnitude. The results reveal substantial heterogeneity in hiring adjustments across roles. Technical positions experience the sharpest declines: a one standard deviation increase in firm-level LLM exposure reduces the likelihood of posting software engineer positions by 1.1 percentage points ($p < 0.001$), QA tester positions by 1.0 percentage points ($p < 0.001$), and data analyst positions by 0.8 percentage points ($p < 0.001$). We find similar declines for other technical roles such as IT specialists, IT project managers, and infrastructure engineers. Creative and design roles also see significant reductions, with job postings for designer positions decreasing by 0.6 percentage points ($p < 0.001$) and for producers by 0.5 percentage points ($p < 0.001$). In contrast, sales representative positions show a statistically significant increase of 0.3 percentage points ($p < 0.001$). This pattern suggests that exposed firms are scaling back recruitment for technical and analytical roles where LLMs can augment or

Figure 5: Effect of LLM exposure on firm-level hiring across job categories



Notes: This figure reports coefficient estimates and 95% confidence intervals for the effect of LLM exposure on firms' hiring at the extensive margin across 50 job roles. For each role, we estimate Equation (4) with firm fixed effects and industry-by-period fixed effects. Standard errors are clustered at the firm level. The dependent variable equals one if a firm posts at least one vacancy for that job role in month t .

replace human capabilities, while maintaining or slightly increasing hiring for roles requiring direct customer interaction and relationship building that remain beyond current LLM capabilities.

Importantly, to emphasize that these estimates capture differential responses between exposed and less exposed firms, not absolute changes in hiring patterns. Hence, a coefficient of zero does not imply that hiring for a particular role remained unchanged. Rather, it indicates that exposed and less exposed firms adjusted their demand for that role similarly. For instance, if all firms reduced hiring of software engineers, regardless of their own exposure levels, our estimates would show no differential effect. The significant negative coefficient for e.g. data analysts therefore reveals that exposed firms reduced analyst hiring beyond any economy-wide reduction, suggesting that firm-level exposure amplifies the substitution effects already occurring across the labor market. We next turn to our occupation-level analysis to examine how hiring intentions of exposed jobs change in the aggregate.

Table 4: Effect of LLM exposure on occupation-level job postings

	Job postings (log)	Salary	Salary (log)	Salary (SD)
LLM exposure x Post	-0.039** (0.018)	-495.483*** (135.270)	-0.011*** (0.003)	-56.802 (108.358)
Occupation fixed-effects	Yes	Yes	Yes	Yes
Year-by-month fixed effects	Yes	Yes	Yes	Yes
Observations	16,815	16,815	16,815	16,804
Adjusted R ²	0.962	0.948	0.953	0.729

Notes: This table reports difference-in-differences estimates of the effect of LLM exposure on occupation-level outcomes. *Job postings (log)* is the log number of new vacancies in occupation o in month t . *Salary* is the mean advertised annual compensation, *Salary (log)* is its logarithm, and *Salary (SD)* is the within-occupation standard deviation of posted salaries. *Exposure* is the occupation-level LLM exposure score from Eloundou et al. (2024). *Post* equals one for November 2022 onward. Standard errors are clustered at the occupation level. Significance is denoted by * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

4.2.2 Occupation-Level Postings

This analysis examines occupation-level job posting data to understand how aggregate labor demand evolves for specific roles. This analysis exploits variation in LLM exposure across 384 occupations, from minimally affected roles such as cooks (exposure score: 0) to highly exposed positions like proofreaders and copy markers (exposure score: 0.86). We examine three outcomes: posting volumes to measure aggregate demand, advertised salaries to assess compensation responses, and within-occupation salary dispersion to measure wage polarization.

Figure 6 presents event study estimates for job posting volumes. Prior to November 2022, high- and low-exposure occupations follow parallel trajectories, validating our identification strategy. Following ChatGPT’s release, however, exposed occupations experience a sustained reduction in hiring demand, with effects first becoming statistically significant after 7 months and intensifying thereafter. After 20 months, we see a slight resurgence, and job postings converge back to their original levels. When averaged across the post-treatment period (Table 4), a one standard deviation increase in LLM exposure is associated with a 3.9% reduction in posting volume ($p = 0.029$). This is an economically meaningful change.

Turning to compensation, we observe immediate and persistent wage pressure for exposed occupations. Figure 7 indicates that posted salaries begin declining shortly after ChatGPT’s introduction. A one standard deviation increase in LLM exposure is associated with a £495 decline in posted salaries and a 1.1% reduction in log